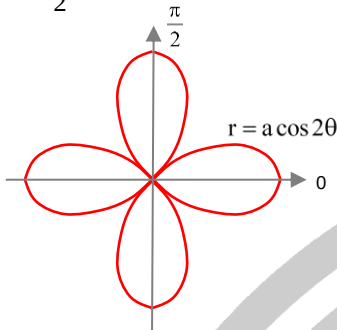




Solutions:

1. A. $\frac{\pi}{2}a^2$

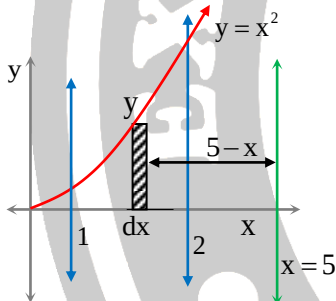


$$A = 8 \int_0^{\pi/4} \frac{1}{2} r^2 d\theta$$

$$A = 4 \int_0^{\pi/4} (a \cos 2\theta)^2 d\theta$$

$$A = 4a^2 \int_0^{\pi/4} (\cos 2\theta)^2 d\theta = \frac{\pi}{2}a^2$$

2. C. 7.92



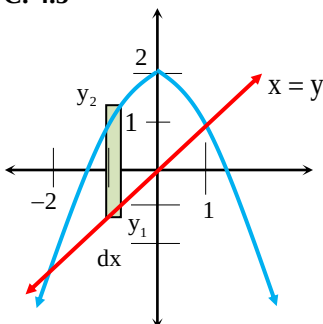
$$M_y = \int (5-x) dA = \int (5-x) y dx$$

$$M_y = \int_1^2 (5-x)(x^2) dx$$

Calculating:

$$M_y = 7.92$$

3. C. 4.5



$$x^2 = -y + 2$$

$$x^2 = -(y-2)$$

vertex : (0, 2)

Point of intersection:

$$x^2 = -y + 2 \text{ and } x = y$$

$$x^2 = -x + 2$$

$$x^2 + x - 2 = 0$$

$$x = 1, y = 1 \text{ ; } x = -2, y = -2$$

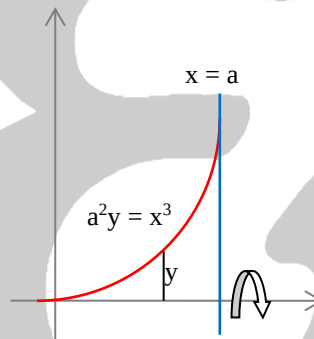
So the points of intersection are (1,1) and (-2, -2)

$$A = \int_{-2}^1 (y_2 - y_1) dx$$

$$A = \int_{-2}^1 (2 - x^2 - x) dx$$

$$A = \frac{9}{2}$$

4. D. $3.56a^2$



$$SA = \int_0^a \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \cdot 2\pi y$$

$$a^2 y = x^3$$

$$a^2 dy = 3x^2 dx$$

$$\frac{dy}{dx} = \frac{3x^2}{a^2}$$

$$y = \frac{x^3}{a^2}$$

$$A = \int_0^a \sqrt{1 + \left(\frac{3x^2}{a^2}\right)^2} dx \cdot 2\pi \left(\frac{x^3}{a^2}\right)$$

$$A = 2\pi \int_0^a \left(\frac{x^3}{a^2}\right) \sqrt{1 + \left(\frac{3x^2}{a^2}\right)^2} dx$$

Calculating:

$$A = 3.56a^2 \rightarrow \text{Ans.}$$

5. A. $-\cos x + \sin x$

Integration by parts

$$\int u dv = uv - \int v du$$

$$u = x$$

$$du = dx$$

$$dv = \sin x$$

$$v = -\cos x$$

$$\begin{aligned} \int x \sin x dx &= -x \cos x + \int \cos x dx \\ &= -x \cos x + \sin x + C \end{aligned}$$

Reverse Engineering:

CALC given:

$$x \sin x \rightarrow \text{CALC} \rightarrow 0.1 \rightarrow = 1.7453 \times 10^{-4}$$

Differentiate choices:

Take choice $-x \cos x + \sin x$

$$\frac{d}{dx} [-x \cos x + \sin x] \Big|_{x=0.1} = 1.7453 \times 10^{-4}$$

6. A. $-\cos x + \frac{1}{3} \cos^3 x + C$

$$\begin{aligned} \int \sin^3 x dx &= \int (\sin x)(\sin^2 x) dx \\ &= \int (\sin x)(1 - \cos^2 x) dx \end{aligned}$$

Let $u = \cos x$

$$du = -\sin x dx$$

$$-du = \sin x dx$$

$$\int \sin^3 x dx = -\int (1 - u^2) du$$

$$\int \sin^3 x dx = -\left(u - \frac{u^3}{3}\right) + c$$

$$\int \sin^3 x dx = -\cos x + \frac{1}{3} \cos^3 x + c$$

Reverse Engineering:

CALC given:

$$(\sin x)^3 \rightarrow \text{CALC} \rightarrow 0.1 \rightarrow = 9.9501 \times 10^{-4}$$

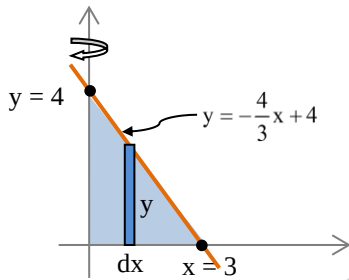
Differentiate choices:

$$-\cos x + \frac{1}{3} \cos^3 x + c$$

$$\frac{d}{dx} \left[-\cos x + \frac{1}{3} \cos^3 x \right] \Big|_{x=0.1} = 9.9501 \times 10^{-4}$$



7. C. 37.7

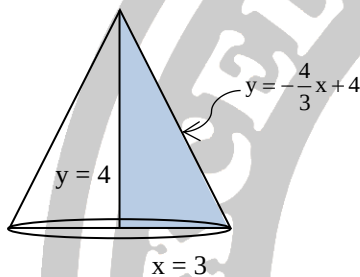


$$V = 2\pi \int_0^3 x \left(-\frac{4}{3}x + 4 \right) dx$$

Calculating:

$$V = 37.7$$

Solution 2:



$$V = \frac{1}{3} Bh$$

$$V = \frac{1}{3} (\pi(3^2)) (4)$$

$$V = 37.7$$

8. A. sec θ + C

$$\int \frac{\sin \theta}{\cos^2 \theta} d\theta = \int \tan \theta \sec \theta d\theta$$

$$= \sec \theta + C$$

Reverse Engineering:

CALC given:

$$\frac{\sin x}{\cos^2 x} \rightarrow \text{CALC} \rightarrow 0.1 \rightarrow = 0.10084$$

Differentiate choices:

Take choice sec θ + C

$$\frac{d}{dx} [\sec x]_{x=0.1} = 0.10084$$

9. B. 0.022

$$\int \frac{x dx}{(x+1)^8} = (x+1)^{-8} x dx$$

Let:

$$u = x$$

$$du = dx$$

$$dv = (x+1)^{-8} dx$$

$$v = \frac{(x+1)^{-7}}{-7}$$

$$\int_0^1 (x+1)^{-8} x dx = uv - \int v du$$

$$= x \left(\frac{(x+1)^{-7}}{-7} \right) - \int \frac{(x+1)^{-7}}{-7} dx$$

$$= \frac{-x}{7(x+1)^7} + \frac{1}{7} \int (x+1)^{-7} dx$$

$$= \frac{-x}{7(x+1)^7} + \frac{1}{7} \left(\frac{(x+1)^{-6}}{-6} \right)$$

$$= \frac{-x}{7(x+1)^7} - \frac{1}{42} \left[(x+1)^{-6} \right]_0^1$$

$$= \frac{-1}{7(1+1)^7} - \frac{0}{7(0+1)^7} - \frac{1}{42} \left[(1+1)^{-6} - (0+1)^{-6} \right]$$

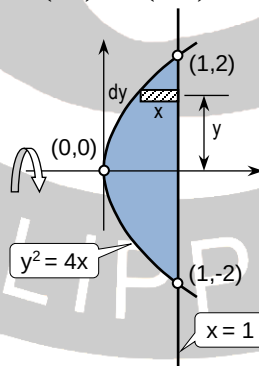
$$= 0.022$$

10. B. 2.13

$$y^2 = 4x$$

$$\text{At } x = 1, x = \pm 2$$

Thus, the parabola and the line intersect at points (1,2) and (1,-2).



$$I_x = 2 \int_0^2 y^2 dA = 2 \int_0^2 y^2 (x dy)$$

$$I_x = 2 \int_0^2 y^2 (x_L - x_P) dy$$

$$I_x = 2 \int_0^2 y^2 \left(1 - \frac{y^2}{4} \right) dy$$

$$I_x = 2 \int_0^2 \left(y^2 - \frac{y^4}{4} \right) dy$$

$$I_x = 2 \left(\frac{y^3}{3} - \frac{y^5}{4(5)} \right) \Big|_0^2$$

$$I_x = 2 \left(\frac{(2)^3}{3} - \frac{(2)^5}{20} \right)$$

$$I_x = 2.13$$

11. A. 0.456

$$\int_5^6 x(x-5)^{12} dx$$

$$\text{Note: } \int u dv = uv - \int v du$$

Let:

$$u = x$$

$$du = dx$$

$$dv = (x-5)^{12} dx$$

$$v = \frac{(x-5)^{13}}{13}$$

$$\int_5^6 x(x-5)^{12} dx = \left[\frac{x(x-5)^{13}}{13} - \frac{1}{13} \int (x-5)^{13} dx \right]_5^6$$

$$= \left[\frac{x}{13} (x-5)^{13} - \frac{1}{13} \left(\frac{(x-5)^{14}}{14} \right) \right]_5^6$$

$$= \left\{ \frac{6}{13} (6-5)^{13} - \frac{5}{13} (5-5)^{13} - \left[\frac{6}{182} [(6-5)^{14} - (5-5)^{14}] \right] \right\}$$

$$\int_5^6 x(x-5)^{12} dx = 0.456$$

12. D. ln 2

$$\int_{-6}^{-10} \frac{dx}{x+2}$$

$$\text{Note: } \int \frac{du}{u} = \ln u$$



$$\begin{aligned}\int_{-6}^{-10} \frac{dx}{x+2} &= \ln(x+2) \Big|_{-6}^{-10} \\ &= \ln(-10+2) - \ln(-6+2) \\ &= \ln(-8) - \ln(-4) \\ &= \ln\left(\frac{-8}{-4}\right) = \ln 2\end{aligned}$$

13. C. 0.0417

$$\int_0^{\pi/2} (\sin^5 x \cos^3 x) dx$$

Using Wallis formula: $m = 5$; $n = 3$; and $\alpha = 1$, since both m & n are odd numbers

$$\begin{aligned}\int_0^{\pi/2} \sin^5 x \cos^3 x dx &= \frac{(5-1)(5-3)(3-1)}{8(6)(4)(2)} \cdot (1) \\ &= 0.0417\end{aligned}$$

14. B. 1

$$\int_1^e \ln x dx$$

Let: $u = \ln x$; $du = \frac{dx}{x}$; $dv = dx$; $v = x$

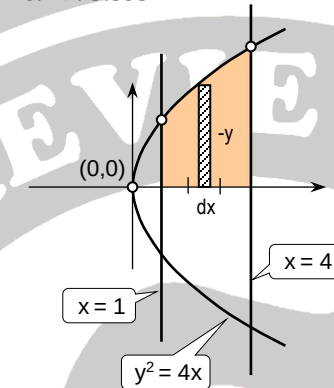
$$\begin{aligned}\int_1^e \ln x dx &= uv - \int v du \\ &= (\ln x)(x) - \int x \left(\frac{dx}{x}\right) \\ &= x \ln x - x \Big|_1^e \\ &= e(\ln e) - e - [1(\ln 1) - 1] \\ &= 1\end{aligned}$$

15. D. 1/3

$$\begin{aligned}\int_0^{\pi} \int_0^{\cos u} r \sin u dr du &= \int_0^{\pi} \frac{1}{2} r^2 \Big|_0^{\cos u} \sin u du \\ &= \frac{1}{2} \int_0^{\pi} (\cos^2 u - 0^2) \sin u du \\ &= -\frac{1}{2} \int_0^{\pi} \cos^2 u (-\sin u) du \\ &= -\frac{1}{2} \frac{\cos^3 u}{3} \Big|_0^{\pi} \\ &= \frac{1}{3}\end{aligned}$$

$$\begin{aligned}&= -\frac{1}{6} (\cos^3 \pi - \cos^3 0) \\ &= -\frac{1}{6} (-1 - 1) \\ &= \frac{1}{3}\end{aligned}$$

16. D. 5.595



$$\begin{aligned}y^2 &= 4x \\ y &= 2\sqrt{x}\end{aligned}$$

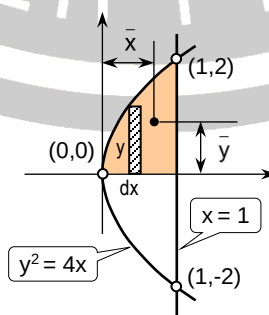
$$\begin{aligned}A &= \int_1^4 y dx = 2 \int_1^4 \sqrt{x} dx \\ &= \frac{2x^{3/2}}{3/2} = \frac{4}{3} x^{3/2} \Big|_1^4 \\ &= \frac{4}{3} [(4)^{3/2} - (1)^{3/2}] \\ &= \frac{4}{3} [8 - 1] \\ &= \frac{4}{3} (7) = \frac{28}{3} \approx 9.33\end{aligned}$$

$A = 5.595$ square units

17. B. (3/5, 3/4)

$$\begin{aligned}y^2 &= 4x \\ \text{at } x = 1, y &= \pm 2\end{aligned}$$

Thus, the parabola and the line intersect at points (1, 2) and (1, -2).



Solving for the area:

$$A = \frac{1}{2} \left(\frac{2}{3} bh \right) = \frac{1}{3} (4)(1) = \frac{4}{3}$$

$$\begin{aligned}A \bullet \bar{x} &= \int_0^1 y dx \bullet x \\ &= \int_0^1 (2\sqrt{x}) dx \bullet x \\ &= 2 \int_0^1 x^{3/2} dx \\ &= 2 \left(\frac{x^{5/2}}{5/2} \right) \Big|_0^1 \\ &= \frac{4}{5} \left(\frac{2}{3} \right) = \frac{4}{5} (1)^{5/2} \\ &= \frac{4}{5} \left(\frac{2}{3} \right) = \frac{4}{5} \left(\frac{2}{3} \right) = \frac{8}{15}\end{aligned}$$

$$= 2 \int_0^1 x^{3/2} dx$$

$$= 2 \left(\frac{x^{5/2}}{5/2} \right) \Big|_0^1$$

$$\frac{4}{3} \left(\frac{2}{3} \right) = \frac{4}{5} (1)^{5/2}$$

$$\bar{x} = \frac{3}{5}$$

$$A \bullet \bar{y} = \int_0^1 y dx \bullet \frac{y}{2} = \frac{1}{2} \int_0^1 y^2 dx$$

$$A \bullet \bar{y} = \frac{1}{2} \int_0^1 (4x) dx$$

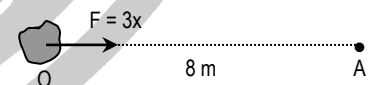
$$= \frac{1}{2} \left(\frac{4x^2}{2} \right) \Big|_0^1$$

$$\frac{4}{3} \left(\frac{2}{3} \right) = (1)^2$$

$$\bar{y} = \frac{3}{4}$$

Thus, the center is at point (3/5, 3/4).

18. C. 96 J



Work = Force x distance

Since the force varies with distance, use integration:

$$W = \int_0^8 F dx$$

$$W = \int_0^8 3x dx$$



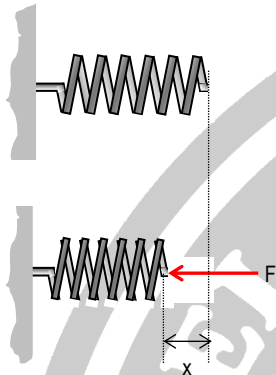
Integrating:

$$W = \frac{3x^2}{2} \Big|_0^8 = \frac{3(8)^2}{2} - \frac{3(0)^2}{2}$$

$$W = 96 \text{ N} \cdot \text{m}$$

$$W = 96 \text{ J}$$

19. A. 3375 N-cm



$$W = \int_3^6 F dx \rightarrow \text{Eq. 1}$$

Using Hooke's law:

$$F = kx$$

$$750 = k(3)$$

$$k = 250$$

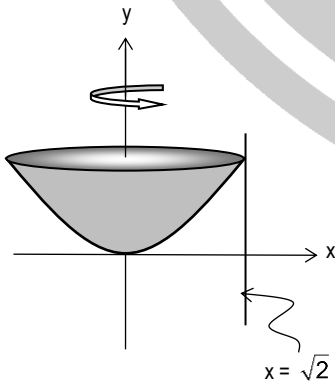
$$F = 250x$$

$$W = \int_3^6 250x dx = 250 \frac{x^2}{2} \Big|_3^6$$

$$W = 125(6^2 - 3^2)$$

$$W = 3375 \text{ N-cm}$$

20. D. 13.61



$$A = \int \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \cdot 2\pi x dx$$

Solving for $\left(\frac{dy}{dx}\right)^2$:

$$y = x^2$$

$$dy = 2x dx$$

$$\frac{dy}{dx} = 2x$$

$$\frac{d^2y}{dx} = 4x^2$$

Substituting in the formula:

$$A = 2\pi \int_0^{\sqrt{2}} \sqrt{1 + 4x^2} x dx$$

$$A = 2\pi \left(\frac{1}{8}\right) \int_0^{\sqrt{2}} (1 + 4x^2)^{\frac{3}{2}} 8x dx$$

$$A = \frac{\pi}{4} \frac{(1 + 4x^2)^{\frac{5}{2}}}{\frac{5}{2}} \Big|_0^{\sqrt{2}}$$

Substituting limits:

$$A = \frac{2\pi}{12} \left\{ [1 + 4(\sqrt{2})^2]^{\frac{5}{2}} - [1 + 4(0)^2]^{\frac{5}{2}} \right\}$$

$$A = 13.61 \text{ sq. units}$$

21. D. 8.17

$$\int_0^2 x \sqrt{4x^2 + 9} dx$$

Let:

$$u = 4x^2 + 9$$

$$du = 8x dx$$

$$\int_0^2 x \sqrt{4x^2 + 9} dx = \frac{1}{8} \int_0^2 \sqrt{4x^2 + 9} (8x dx)$$

$$\text{But } \int u^{\frac{1}{2}} du = \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$\int_0^2 x \sqrt{4x^2 + 9} dx = \frac{1}{8} \frac{(4x^2 + 9)^{\frac{3}{2}}}{\frac{3}{2}} \Big|_0^2$$

$$\int_0^2 x \sqrt{4x^2 + 9} dx = \frac{1}{12} \left\{ [4(2)^2 + 9]^{\frac{3}{2}} - [4(0)^2 + 9]^{\frac{3}{2}} \right\}$$

$$= 8.17$$

22. B. 21.055 kJ

After lifting the load x meters, the total mass of the cable left in the shaft as well as the load is:

$$\frac{1}{2}(50 - x) + 150 = 175 - \frac{x}{2}$$

Note: Weight in kg is:

$$\text{Weight} = \text{mass}(g) \quad g = 9.8 \text{ m/s}^2$$

The force required to hold the cable and load a distance of x meters above the bottom is:

$$F(x) = 9.8 \left(175 - \frac{x}{2} \right)$$

$$F(x) = 1715 - 4.9x$$

$$W = \int_0^{12.5} (1715 - 4.9x) dx$$

$$W = [1715x - 2.45x^2]_0^{12.5}$$

$$W = 21054.69 \text{ J} = 21.055 \text{ kJ}$$

23. D. 15

$$\iiint_B 8xyz dV = \int_1^2 \int_2^3 \int_0^1 8xyz dz dx dy$$

$$= \int_1^2 \int_2^3 4xy \Big|_0^1 dx dy = \int_1^2 4xy dx dy$$

$$= \int_1^2 2x^2 y \Big|_2^3 dy = \int_1^2 10y dy = 15$$

24. B. 4π

The cross sections are disks with radius r

$$= y \text{ where } y = \frac{1}{3}\sqrt{9 - x^2}$$

$$V = \pi \int_{-3}^3 y^2 dx = 2\pi \int_0^3 \frac{1}{9}(9 - x^2) dx$$

$$V = \frac{2\pi}{9} \left(9x - \frac{1}{3}x^3 \right) \Big|_0^3 = 4\pi$$

25. D. $\frac{1}{2} \ln 3$

$$\text{ave value} = \frac{1}{2} \int_1^2 \frac{1}{x} dx$$

$$= \frac{1}{2} \ln x \Big|_1^2 = \frac{1}{2} (\ln 2 - \ln 1) = \frac{1}{2} \ln 2$$

END